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Inventor(s) BALLARD; Adam Richard et al.

INVERTED BATTERY ENCLOSURE

Abstract

Aspects of the subject technology relate to an inverted battery enclosure, such as an inverted battery enclosure for a battery pack for a vehicle. The inverted battery enclosure may include a lid, and one or more battery cells and/or battery subassemblies attached to an interior surface of the lid. A gap may be provided, within the enclosure, between the battery cells and a bottom of the enclosure. One or more impact absorption features may be provided along a sidewall of the lid.

Inventors: **BALLARD; Adam Richard (Corona, CA), KHATER; Piyush (Mission Viejo, CA), JACOBS; Tyler (Hawthorne, CA), GANAPATHY; Mukund (Milpitas, CA), ACHAR; Sriniket Sreenivas (Sterling Heights, MI), JOHANSON; Stephen John (New Malden, GB), WILSON; Jonathan Christopher (Rancho Mission Viejo, CA), SONAWANE; Atul Arun (Irvine, CA), LIM; Youngbin (Irvine, CA), THOMBRE; Akash (Tustin, CA)**

Applicant: **Rivian IP Holdings, LLC** (Irvine, CA)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/557,395, entitled, “Inverted Battery Enclosure”, filed on Feb. 23, 2024, the disclosure of which is hereby incorporated herein in its entirety.

INTRODUCTION

[0002] Batteries are often used as a source of power, including as a source of power for electric vehicles that include wheels that are driven by an electric motor that receives power from the battery.

[0003] Aspects of the subject technology can help to improve the efficiency, serviceability, reliability, and/or range of electric vehicles, which can help to mitigate climate change by reducing greenhouse gas emissions.

SUMMARY

[0004] Aspects of the subject technology relate to an inverted battery assembly for a battery pack, such as a battery pack for a vehicle. For example, an inverted battery assembly may form an energy volume for the battery pack. The inverted battery assembly may include a lid, and one or more battery cells, battery modules, and/or other battery subassemblies attached to the lid. In this way, the torsional stiffness of the battery pack, and/or a vehicle in which the battery pack is installed, may be increased. Modal performance of the battery pack, and/or a vehicle in which the battery pack is installed, may also be improved. In one or more implementations, the lid of the battery pack may form a floor of a vehicle. This can reduce the amount of material used to form the vehicle, which can improve the efficiency, and/or range of electric vehicles. The lid may be provided with sidewall structures for improving side impact performance of the battery pack, and/or a vehicle in which the battery pack is installed.

[0005] In accordance with aspects of the disclosure, an apparatus is provided that includes a battery pack, the battery pack including a lid that forms a top and a plurality of sidewalls of an energy volume enclosure of the battery pack; one or more battery subassemblies enclosed within a space defined by the top and the plurality of sidewalls of the lid; and a tray attached to the lid to enclose the one or more battery subassemblies within the space defined by the top and the plurality of sidewalls of the lid. The tray may form a bottom wall of the energy volume enclosure and may be configured to attach to the lid using a plurality of fasteners disposed around a periphery of the energy volume enclosure to enclose the one or more battery subassemblies within the space defined by the top and the plurality of sidewalls of the lid. The battery pack may also include an air gap between the tray and the one or more battery subassemblies.

[0006] The one or more battery subassemblies may each include one or more battery cells, each of the one or more battery cells having a vent disposed adjacent the air gap. Each of the one or more battery subassemblies may be attached to the lid. Each of the one or more battery subassemblies may include a current collector assembly having a standoff that defines a spacing between the one or more battery subassemblies and the lid. The battery pack may also include a potting material in an interstitial space between two or more battery cells of each of the one or more battery subassemblies, and in a space between the one or more battery subassemblies and the lid. The air gap between the tray and the one or more battery subassemblies may be free of the potting material, and the potting material may attach the one or more battery subassemblies to the lid.

[0007] The tray may include a plurality of ribs; and a feature between adjacent pairs of the plurality of ribs, the feature configured to interface with a longitudinal member of the energy volume enclosure to fluidly isolate a first region within the energy volume enclosure from a second region within the energy volume enclosure. The battery pack may also include a skid plate attached to an outer surface of the tray. The tray may be attached to the longitudinal member of the energy volume enclosure by one or more fasteners, and the skid plate may include one or more ribs positioned within a space defined by one of the ribs of the tray. The apparatus may also include a plurality of impact absorption features mounted along one of the plurality of sidewalls of the lid. The apparatus may include a vehicle, and the battery pack may be implemented in the vehicle.
[0008] In accordance with other aspects of the disclosure, an enclosure for a battery pack may be provided, the enclosure including a front end; a rear end; a sidewall extending between the front end and the rear end; a first plurality of impact absorption features on an external surface of the sidewall; and a second plurality of impact absorption features on an internal surface of the sidewall. The enclosure may also include a lid; and an outer longitudinal member extending from the front end to the rear end. The second plurality of impact absorption features may include a plurality of bulkheads that are attached to the outer longitudinal member and positioned interior to the lid. The first plurality of impact absorption features may include at least one beam attached to an outer surface of the lid, the at least one beam extending in a direction that runs between the front end and the rear end. The at least one beam may include a first beam attached to the outer surface of the lid at a top of the sidewall; and a second beam attached to a flange at a bottom of the lid. The first plurality of impact absorption features may include a plurality of reinforcement members attached to, and spaced apart along, the second beam.
[0009] In accordance with other aspects of the disclosure, a method may be provided that includes lowering a battery subassembly into a lid of an energy volume enclosure of a battery pack while the lid is in an inverted position; attaching the battery subassembly to the lid; providing a potting material into an interstitial space between two or more battery cells in the battery subassembly; flipping the lid having the battery subassembly attached thereto; and enclosing the battery subassembly within the energy volume enclosure by attaching a tray of the energy volume enclosure to the lid of the energy volume enclosure. Lowering the battery subassembly into the lid may include lowering the battery subassembly into the lid while the battery subassembly is in an inverted configuration of the battery subassembly. The method may also include providing the battery pack in a vehicle. Attaching the battery subassembly to the lid may include attaching the battery subassembly to the lid with the potting material. The method may also include mechanically coupling a modular electrical component assembly to the lid, and electrically coupling the modular electrical component assembly to the battery subassembly via the lid.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0010] Certain features of the subject technology are set forth in the appended claims. However, for purpose of explanation, several embodiments of the subject technology are set forth in the following figures.
[0011] FIGS. **1**A and **1**B illustrate schematic perspective side views of example implementations of a vehicle having a battery pack in accordance with one or more implementations.
[0012] FIG. **1**C illustrates a schematic perspective view of a building having a battery pack in accordance with one or more implementations.
[0013] FIG. **2**A illustrates a schematic perspective view of a battery pack in accordance with one or more implementations.
[0014] FIG. **2**B illustrates schematic perspective views of various battery modules that may be

included in a battery pack in accordance with one or more implementations.
[0015] FIG. **2**C illustrates a cross-sectional end view of a battery cell in accordance with one or more implementations.
[0016] FIG. **2**D illustrates a cross-sectional perspective view of a cylindrical battery cell in accordance with one or more implementations.
[0017] FIG. **2**E illustrates a cross-sectional perspective view of a prismatic battery cell in accordance with one or more implementations.
[0018] FIG. **2**F illustrates a cross-sectional perspective view of a pouch battery cell in accordance with one or more implementations.
[0019] FIG. **3** illustrates a cross-sectional side view of a battery pack in accordance with one or more implementations.
[0020] FIG. **4** illustrates a battery pack in various stages of assembly in accordance with one or more implementations.
[0021] FIG. **5** illustrates a partially exploded rear view of a battery pack in accordance with one or more implementations.
[0022] FIG. **6** illustrates an assembled rear view of a battery pack in accordance with one or more implementations.
[0023] FIG. **7** illustrates a top view of a battery subassembly in accordance with one or more implementations.
[0024] FIG. **8**A illustrates a top view of an inverted lid for a battery pack with a battery subassembly installed therein in accordance with one or more implementations.
[0025] FIG. **8**B illustrates a cross-sectional side view of a process for providing a battery subassembly in an inverted lid of a battery pack in accordance with one or more implementations.
[0026] FIG. **9** illustrates a cross-sectional view of a portion of a battery pack in an inverted configuration during assembly in accordance with one or more implementations.
[0027] FIG. **10** illustrates aspects of datuming for a battery pack assembly in accordance with one or more implementations.
[0028] FIG. **11** illustrates a top view of a tray for a battery pack in accordance with one or more implementations.
[0029] FIG. **12** illustrates a cross-sectional view of a portion of a battery pack in accordance with one or more implementations.
[0030] FIG. **13** illustrates a cross-sectional view of another portion of a battery pack in accordance with one or more implementations.
[0031] FIG. **14** illustrates a perspective view of a portion of a sidewall of a battery pack in accordance with one or more implementations.
[0032] FIG. **15** illustrates a cross-sectional side view of a portion of a sidewall of battery pack in accordance with one or more implementations.
[0033] FIG. **16** illustrates a perspective view of an outer longitudinal member of a battery pack in accordance with one or more implementations.
[0034] FIG. **17** illustrates a perspective view of beam of a battery pack in accordance with one or more implementations.
[0035] FIG. **18** illustrates aspects of a battery pack having impact absorption features implemented in a vehicle, in accordance with one or more implementations.
[0036] FIG. **19** illustrates a flow chart of illustrative operations that may be performed for assembling a battery pack in accordance with one or more implementations.

DETAILED DESCRIPTION

[0037] The detailed description set forth below is intended as a description of various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology can be practiced. The appended drawings are incorporated herein and constitute a part of the detailed description. The detailed description includes specific details for the purpose of providing a thorough understanding of the subject technology. However, the subject technology is not limited to the specific details set forth herein and can be practiced using one or more other implementations. In one or more implementations, structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.

[0038] Aspects of the subject technology described herein relate to an inverted battery assembly. The inverted battery assembly may include one or more impact absorption features along a sidewall of the inverted battery assembly.

[0039] FIG. **1**A is a diagram illustrating an example implementation of a moveable apparatus as described herein. In the example of FIG. **1**A, a moveable apparatus is implemented as a vehicle **100**. As shown, the vehicle **100** may include one or more battery packs, such as battery pack **110**. The battery pack **110** may be coupled to one or more electrical systems of the vehicle **100** to provide power to the electrical systems.

[0040] In one or more implementations, the vehicle **100** may be an electric vehicle having one or more electric motors that drive the wheels **102** of the vehicle using electric power from the battery pack **110**. In one or more implementations, the vehicle **100** may also, or alternatively, include one or more chemically powered engines, such as a gas-powered engine or a fuel cell powered motor. For example, electric vehicles can be fully electric or partially electric (e.g., hybrid or plug-in hybrid.

[0041] In the example of FIG. **1**A, the vehicle **100** is implemented as a truck (e.g., a pickup truck) having a battery pack **110**. As shown, the battery pack **110** may include one or more battery modules **115**, which may include one or more battery cells **120**. As shown in FIG. **1**A, the battery pack **110** may also, or alternatively, include one or more battery cells **120** mounted directly in the battery pack **110** (e.g., in a cell-to-pack configuration). In one or more implementations, the battery pack **110** may be provided without any battery modules **115** and with the battery cells **120** mounted directly in the battery pack **110** (e.g., in a cell-to-pack configuration) and/or in other battery units that are installed in the battery pack **110**. A vehicle battery pack can include multiple energy storage devices that can be arranged into such as battery modules or battery units. A battery unit or module can include an assembly of cells that can be combined with other elements (e.g., structural frame, thermal management devices) that can protect the assembly of cells from heat, shock and/or vibrations.

[0042] For example, the battery cell **120** can be included a battery, a battery unit, a battery module and/or a battery pack to power components of the vehicle **100**. For example, a battery cell housing of the battery cell **120** can be disposed in the battery module **115**, the battery pack **110**, a battery array, or other battery unit installed in the vehicle **100**.

[0043] As discussed in further detail hereinafter, the battery cells **120** may be provided with a battery cell housing that can be provided with any of various outer shapes. The battery cell housing may be a rigid housing in some implementations (e.g., for cylindrical or prismatic battery cells). The battery cell housing may also, or alternatively, be formed as a pouch or other flexible or malleable housing for the battery cell in some implementations. In various other implementations, the battery cell housing can be provided with any other suitable outer shape, such as a triangular outer shape, a square outer shape, a rectangular outer shape, a pentagonal outer shape, a hexagonal outer shape, or any other suitable outer shape. In some implementations, the battery pack **110** may not include modules (e.g., the battery pack may be module-free). For example, the battery pack **110** can have a module-free or cell-to-pack configuration in which the battery cells **120** are arranged directly into the battery pack **110** without assembly into a battery module **115**. In one or more implementations, the vehicle **100** may include one or more busbars, electrical connectors, or other charge collecting, current collecting, and/or coupling components to provide electrical power from the battery pack **110** to various systems or components of the vehicle **100**. In one or more implementations, the vehicle **100** may include control circuitry such as a power stage circuit that can be used to convert DC power from the battery pack **110** into AC power for one or more

components and/or systems of the vehicle (e.g., including one or more power outlets of the vehicle and/or the motor(s) that drive the wheels **102** of the vehicle). The power stage circuit can be provided as part of the battery pack **110** or separately from the battery pack **110** within the vehicle **100**. The vehicle **100** may have a front end **131** and a rear end **133**.
[0044] The example of FIG. **1**A in which the vehicle **100** is implemented as a pickup truck having a truck bed at the rear portion thereof is merely illustrative. For example, FIG. **1**B illustrates another implementation in which the vehicle **100** including the battery pack **110** is implemented as a sport utility vehicle (SUV), such as an electric sport utility vehicle. In the example of FIG. **1**B, the vehicle **100** including the battery pack **110** may include a cargo storage area that is enclosed within the vehicle **100** (e.g., behind a row of seats within a cabin of the vehicle). In other implementations, the vehicle **100** may be implemented as another type of electric truck, an electric delivery van, an electric automobile, an electric car, an electric motorcycle, an electric scooter, an electric bicycle, an electric passenger vehicle, an electric passenger or commercial truck, a hybrid vehicle, an aircraft, a watercraft, and/or any other movable apparatus having a battery pack **110** (e.g., a battery pack or other battery unit that powers the propulsion or drive components of the moveable apparatus).
[0045] In one or more implementations, a battery pack such as the battery pack **110**, a battery module **115**, a battery cell **120**, and/or any other battery unit as described herein may also, or alternatively, be implemented as an electrical power supply and/or energy storage system in a building, such as a residential home or commercial building. For example, FIG. **1**C illustrates an example in which a battery pack **110** is implemented in a building **180**. For example, the building **180** may be a residential building, a commercial building, or any other building. As shown, in one or more implementations, a battery pack **110** may be mounted to a wall of the building **180**.
[0046] As shown, the battery **110**A that is installed in the building **180** may be couplable to the battery pack **110** in the vehicle **100**, such as via: a cable/connector **106** that can be connected to the charging port **130** of the vehicle **100**, electric vehicle supply equipment **170** (EVSE), a power stage circuit **172**, and/or a cable/connector **174**. For example, the cable/connector **106** may be coupled to the EVSE **170**, which may be coupled to the battery **110**A via the power stage circuit **172**, and/or may be coupled to an external power source **190**. In this way, either the external power source **190** or the battery **110**A that is installed in the building **180** may be used as an external power source to charge the battery pack **110** in the vehicle **100** in some use cases. In some examples, the battery **110**A that is installed in the building **180** may also, or alternatively, be coupled (e.g., via a cable/connector **174**, the power stage circuit **172**, and the EVSE **170**) to the external power source **190**. For example, the external power source **190** may be a solar power source, a wind power source, and/or an electrical grid of a city, town, or other geographic region (e.g., electrical grid that is powered by a remote power plant). During, for example, times when the battery pack **110** in the vehicle **100** is not coupled to the battery **110**A that is installed in the building **180**, the battery **110**A that is installed in the building **180** can be coupled (e.g., using the power stage circuit **172** for the building **180**) to the external power source **190** to charge up and store electrical energy. In some use cases, this stored electrical energy in the battery **110**A that is installed in the building **180** can later be used to charge the battery pack **110** in the vehicle **100** (e.g., during times when solar power or wind power is not available, in the case of a regional or local power outage for the building **180**, and/or during a period of high rates for access to the electrical grid).
[0047] In one or more implementations, the power stage circuit **172** may electrically couple the battery **110**A that is installed in the building **180** to an electrical system of the building **180**. For example, the power stage circuit **172** may convert DC power from the battery **110**A into AC power for one or more loads in the building **180**. For example, the battery **110**A that is installed in the building **180** may be used to power one or more lights, lamps, appliances, fans, heaters, air conditioners, and/or any other electrical components or electrical loads in the building **180** (e.g., via one or more electrical outlets that are coupled to the battery **110**A that is installed in the building

180). For example, the power stage circuit **172** may include control circuitry that is operable to switchably couple the battery **110**A between the external power source **190** and one or more electrical outlets and/or other electrical loads in the electrical system of the building **180**. In one or more implementations, the vehicle **100** may include a power stage circuit (not shown in FIG. **1**C) that can be used to convert power received from the electric vehicle supply equipment **170** to DC power that is used to power/charge the battery pack **110** of the vehicle **100**, and/or to convert DC power from the battery pack **110** into AC power for one or more electrical systems, components, and/or loads of the vehicle **100**.

[0048] In one or more use cases, the battery **110**A that is installed in the building **180** may be used as a source of electrical power for the building **180**, such as during times when solar power or wind power is not available, in the case of a regional or local power outage for the building **180**, and/or during a period of high rates for access to the electrical grid (as examples). In one or more other use cases, the battery pack **110** that is installed in the vehicle may be used to charge the battery **110**A that is installed in the building **180** and/or to power the electrical system of the building **180** (e.g., in a use case in which the battery **110**A that is installed in the building **180** is low on or out of stored energy and in which solar power or wind power is not available, a regional or local power outage occurs for the building **180**, and/or a period of high rates for access to the electrical grid occurs (as examples)).

[0049] FIG. **2**A depicts an example battery pack **110**, in accordance with one or more implementations. As shown, the battery pack **110** may include an energy volume enclosure **205** (e.g., a battery pack housing, sometimes referred to herein as an enclosure). For example, the energy volume enclosure **205** may house or enclose an energy volume **207** for the battery pack **110**, the energy volume **207** including one or more battery modules **115** and/or one or more battery cells **120**, and/or other battery pack components. In one or more implementations, the energy volume enclosure **205** may include or form a shielding structure on an outer surface thereof (e.g., a bottom thereof and/or underneath one or more battery module **115**, battery units, batteries, and/or battery cells **120**) to protect the battery module **115**, battery units, batteries, and/or battery cells **120** from external conditions (e.g., if the battery pack **110** is installed in a vehicle **100** and the vehicle **100** is driven over rough terrain, such as off-road terrain, trenches, rocks, rivers, streams, etc.).

[0050] Battery pack **110** may include, within the energy volume **207** and the energy volume enclosure **205**, multiple battery cells **120** (e.g., directly installed within the battery pack **110**, or within batteries, battery units, battery subassemblies, and/or battery modules **115** as described herein) and/or battery modules **115**, and one or more conductive coupling elements for coupling a voltage generated by the battery cells **120** to a power-consuming component, such as the vehicle **100** and/or an electrical system of a building **180**. For example, the conductive coupling elements may include internal connectors and/or contactors that couple together multiple battery cells **120**, battery units, batteries, battery subassemblies, and/or multiple battery modules **115** within the energy volume enclosure **205** to generate a desired output voltage for the battery pack **110**.

[0051] As shown, the battery pack **110** may also include a modular electrical component assembly **290** (e.g., including a modular electronic component enclosure or a modular electrical component enclosure) mounted to the energy volume enclosure **205**. In one or more implementations, the modular electrical component assembly **290** may include one or more of the conductive coupling elements for routing power from the battery cells **120** and/or battery modules **115** within the energy volume enclosure **205** (e.g., within the energy volume **207**) to one or more external connection ports, such as an electrical contact **203** (e.g., a high voltage terminal, port, or connector). For example, an electrical cable or harness may be connected between the electrical contact **203** and an electrical system of the vehicle **100** or the building **180**, to provide electrical power to the vehicle **100** or the building **180**. The energy volume enclosure **205** may have a front end **267** and a rear end **269**. In one or more implementations, when the battery pack **110** is installed in the vehicle **100**, the battery pack **110** may be arranged with the front end **267** closer to the front end **131** of the vehicle

and the rear end **269** closer to the rear end **133** of the vehicle. As shown, the modular electrical component assembly **290** may be mounted to the energy volume enclosure **205** (e.g., to a lid **277** of the energy volume enclosure **205**) at or near the rear end **269** in one or more implementations.
[0052] In one or more implementations, the battery pack **110** may include one or more additional features, such as thermal control structures (e.g., cooling lines and/or plates and/or heating lines and/or plates). For example, thermal control structures may couple thermal control structures and/or fluids to the battery modules **115**, battery units, batteries, and/or battery cells **120** within the energy volume enclosure **205**, such as by distributing fluid through the battery pack **110**.
[0053] For example, the thermal control structures may form a part of a thermal/temperature control or heat exchange system that includes one or more thermal components such as plates or bladders that are disposed in thermal contact with one or more battery modules **115** and/or battery cells **120** disposed within the energy volume enclosure **205**. For example, a thermal component may be positioned in contact with one or more battery modules **115**, battery units, batteries, and/or battery cells **120** within the energy volume enclosure **205**. In one or more implementations, the battery pack **110** may include one or multiple thermal control structures and/or other thermal components for each of several top and bottom battery module pairs. As shown, the battery pack **110** may include an electrical contact **203** (e.g., a high voltage connector or port) by which an external load (e.g., the vehicle **100** or an electrical system of the building **180**) may be electrically coupled to the battery modules and/or battery cells in the battery pack **110**.
[0054] As shown, the energy volume enclosure **205** of the battery pack **110** may include a lid **277**. For example, the lid **277** may cover and extend over one or more battery modules **115**, battery cells **120**, and/or other battery subassemblies within the energy volume enclosure **205**. In the example of FIG. **2**A, the lid **277** may be a deep-drawn structure that forms a top **257**, and one or more sidewalls **259** (e.g., four sidewalls), of the energy volume enclosure **205**. As discussed in further detail hereinafter, the energy volume enclosure **205** may also include a tray or other housing structure (e.g., at the bottom of the energy volume enclosure) that interfaces with the lid **277** to enclose one or more battery modules **115**, battery cells **120**, and/or other battery subassemblies within the energy volume enclosure **205** (e.g., within a space defined by the top **257** and the sidewalls **259** of the lid **277**). For example, the energy volume enclosure **205** may include a tray panel that is removable to expose an opening in the bottom of the lid **277**.
[0055] In the example of FIG. **2**A, the lid **277** is provided with ribbing **275** (e.g., for additional strength). In the example of FIG. **2**A, the battery pack **110** includes one or more mounting features **273** (e.g., for mounting the battery pack **110** to one or more body structures of a vehicle, such as the vehicle **100**). As shown in FIG. **2**A, and as discussed in further detail hereinafter, the energy volume enclosure **205** may include one or more sidewall structures **271**. The sidewall structures **271** may be attached to, and/or extend long, a sidewall **259** of the lid **277**, and may provide impact absorption and/or redistribution functions to distribute energy from a side impact to the battery pack **110** (e.g., from a side impact to a vehicle **100**) away from and/or around the one or more battery modules **115**, battery cells **120**, and/or other battery subassemblies within the energy volume enclosure **205**.
[0056] FIG. **2**B depicts various examples of battery modules **115** that may be disposed in the battery pack **110** (e.g., within the energy volume enclosure **205** of FIG. **2**A). In the example of FIG. **2**B, a battery module **115**A is shown that includes a battery module housing **223** having a rectangular cuboid shape with a length that is substantially similar to its width. In this example, the battery module **115**A includes multiple battery cells **120** implemented as cylindrical battery cells. In this example, the battery module **115**A includes rows and columns of cylindrical battery cells that are coupled together by an interconnect structure **200** (e.g., a current connector assembly or CCA). For example, the interconnect structure **200** may couple together the positive terminals of the battery cells **120**, and/or couple together the negative battery terminals of the battery cells **120**. As shown, the battery module **115**A may include a charge collector or bus bar **202**. For example,

the bus bar **202** may be electrically coupled to the interconnect structure **200** to collect the charge generated by the battery cells **120** to provide a high voltage output from the battery module **115**A.

[0057] FIG. **2**B also shows a battery module **115**B having an elongate shape, in which the length of the battery module housing **223** (e.g., extending along a direction from a front end of the battery pack **110** to a rear end of the battery pack **110** when the battery module **115**B is installed in the battery pack **110**) is substantially greater than a width (e.g., in a transverse direction to the direction from the front end of the battery pack **110** to the rear end of the battery pack **110** when the battery module **115**B is installed in the battery pack **110**) of the battery module housing **223**. For example, one or more battery modules **115**B may span the entire front-to-back length of a battery pack within the energy volume enclosure **205**. As shown, the battery module **115**B may also include a bus bar **202** electrically coupled to the interconnect structure **200**. For example, the bus bar **202** may be electrically coupled to the interconnect structure **200** to collect the charge generated by the battery cells **120** to provide a high voltage output from the battery module **115**B.

[0058] In the implementations of battery module **115**A and battery module **115**B, the battery cells **120** are implemented as cylindrical battery cells. However, in other implementations, a battery module may include battery cells having other form factors, such as a battery cells having a right prismatic outer shape (e.g., a prismatic cell), or a pouch cell implementation of a battery cell. As an example, FIG. **2**B also shows a battery module **115**C having a battery module housing **223** having a rectangular cuboid shape with a length that is substantially similar to its width and including multiple battery cells **120** implemented as prismatic battery cells. In this example, the battery module **115**C includes rows and columns of prismatic battery cells that are coupled together by an interconnect structure **200** (e.g., a current collector assembly or CCA). For example, the interconnect structure **200** may couple together the positive terminals of the battery cells **120** and/or couple together the negative battery terminals of the battery cells **120**. As shown, the battery module **115**C may include a charge collector or bus bar **202**. For example, the bus bar **202** may be electrically coupled to the interconnect structure **200** to collect the charge generated by the battery cells **120** to provide a high voltage output from the battery module **115**C.

[0059] FIG. **2**B also shows a battery module **115**D including prismatic battery cells and having an elongate shape, in which the length of the battery module housing **223** (e.g., extending along a direction from a front end of the battery pack **110** to a rear end of the battery pack **110** when the battery module **115**D is installed in the battery pack **110**) is substantially greater than a width (e.g., in a transverse direction to the direction from the front end of the battery pack **110** to the rear end of the battery pack **110** when the battery module **115**D is installed in the battery pack **110**) of the battery module housing **223**. For example, one or more battery modules **115**D having prismatic battery cells may span the entire front-to-back length of a battery pack within the energy volume enclosure **205**. As shown, the battery module **115**D may also include a bus bar **202** electrically coupled to the interconnect structure **200**. For example, the bus bar **202** may be electrically coupled to the interconnect structure **200** to collect the charge generated by the battery cells **120** to provide a high voltage output from the battery module **115**D.

[0060] As another example, FIG. **2**B also shows a battery module **115**E having a battery module housing **223** having a rectangular cuboid shape with a length that is substantially similar to its width and including multiple battery cells **120** implemented as pouch battery cells. In this example, the battery module **115**C includes rows and columns of pouch battery cells that are coupled together by an interconnect structure **200** (e.g., a current collector assembly or CCA). For example, the interconnect structure **200** may couple together the positive terminals of the battery cells **120** and couple together the negative battery terminals of the battery cells **120**. As shown, the battery module **115**E may include a charge collector or bus bar **202**. For example, the bus bar **202** may be electrically coupled to the interconnect structure **200** to collect the charge generated by the battery cells **120** to provide a high voltage output from the battery module **115**E.

[0061] FIG. **2**B also shows a battery module **115**F including pouch battery cells and having an elongate shape in which the length of the battery module housing **223** (e.g., extending along a direction from a front end of the battery pack **110** to a rear end of the battery pack **110** when the battery module **115**E is installed in the battery pack **110**) is substantially greater than a width (e.g., in a transverse direction to the direction from the front end of the battery pack **110** to the rear end of the battery pack **110** when the battery module **115**E is installed in the battery pack **110**) of the battery module housing **223**. For example, one or more battery modules **115**E having pouch battery cells may span the entire front-to-back length of a battery pack within the energy volume enclosure **205**. As shown, the battery module **115**E may also include a bus bar **202** electrically coupled to the interconnect structure **200**. For example, the bus bar **202** may be electrically coupled to the interconnect structure **200** to collect the charge generated by the battery cells **120** to provide a high voltage output from the battery module **115**E.

[0062] In various implementations, a battery pack **110** may be provided with one or more of any of the battery modules **115**A, **115**B, **115**C, **115**D, **115**E, and **115**F. In one or more other implementations, a battery pack **110** may be provided without battery modules **115** (e.g., in a cell-to-pack implementation). In one or more implementations, a battery pack **110** may be provided with three elongated battery modules (e.g., three of battery modules **115**B, **115**D, and/or **115**F).

[0063] In one or more implementations, multiple battery modules **115** in any of the implementations of FIG. **2**B may be coupled (e.g., in series) to a current collector of the battery pack **110**. In one or more implementations, the current collector may be coupled, via a high voltage harness, to one or more external connectors (e.g., electrical contact **203**) on the battery pack **110**. In one or more implementations, the battery pack **110** may be provided without any battery modules **115**. For example, the battery pack **110** may have a cell-to-pack configuration in which battery cells **120** are arranged directly into the battery pack **110** without assembly into a battery module **115** (e.g., without including a separate battery module housing **223**). For example, the battery pack **110** (e.g., the energy volume enclosure **205**) may include or define a plurality of structures for positioning of the battery cells **120** directly within the energy volume enclosure **205**.

[0064] FIG. **2**C illustrates a cross-sectional end view of a portion of a battery cell **120**. As shown in FIG. **2**C, a battery cell **120** may include an anode **208**, an electrolyte **210**, and a cathode **212**. As shown, the anode **208** may include or be electrically coupled to a first current collector **206** (e.g., a metal layer such as a layer of copper foil or other metal foil). As shown, the cathode **212** may include or be electrically coupled to a second current collector **214** (e.g., a metal layer such as a layer of aluminum foil or other metal foil). As shown, the battery cell **120** may include a first terminal **216** (e.g., a negative terminal) coupled to the anode **208** (e.g., via the first current collector **206**) and a second terminal **218** (e.g., a positive terminal) coupled to the cathode (e.g., via the second current collector **214**). In various implementations, the electrolyte **210** may be a liquid electrolyte layer or a solid electrolyte layer. In one or more implementations (e.g., implementations in which the electrolyte **210** is a liquid electrolyte layer), the battery cell **120** may include a separator layer **220** that separates the anode **208** from the cathode **212**. In one or more implementations in which the electrolyte **210** is a solid electrolyte layer, the solid electrolyte layer may act as both separator layer and an electrolyte layer.

[0065] In one or more implementations, the battery cell **120** may be implemented as a lithium ion battery cell in which the anode **208** is formed from a carbonaceous material (e.g., graphite or silicon-carbon). In these implementations, lithium ions can move from the anode **208**, through the electrolyte **210**, to the cathode **212** during discharge of the battery cell **120** (e.g., and through the electrolyte **210** from the cathode **212** to the anode **208** during charging of the battery cell **120**). For example, the anode **208** may be formed from a graphite material that is coated on a copper foil corresponding to the first current collector **206**. In these lithium ion implementations, the cathode **212** may be formed from one or more metal oxides (e.g., a lithium cobalt oxide, a lithium manganese oxide, a lithium nickel manganese cobalt oxide (NMC), or the like) and/or a lithium iron phosphate. As shown, the battery cell **120** may include a separator layer **220** that separates the anode **208** from the cathode **212**. In an implementation in which the battery cell **120** is implemented as a lithium-ion battery cell, the electrolyte **210** may include a lithium salt in an organic solvent. The separator layer **220** may be formed from one or more insulating materials (e.g., a polymer such as polyethylene, polypropylene, polyolefin, and/or polyamide, or other insulating materials such as rubber, glass, cellulose or the like). The separator layer **220** may prevent contact between the anode **208** and the cathode **212**, and may be permeable to the electrolyte **210** and/or ions within the electrolyte **210**. In one or more implementations, the battery cell **120** may be implemented as a lithium polymer battery cell having a dry solid polymer electrolyte and/or a gel polymer electrolyte.

[0066] Although some examples are described herein in which the battery cells **120** are implemented as lithium-ion battery cells, some or all of the battery cells **120** in a battery module **115**, battery pack **110**, or other battery or battery unit may be implemented using other battery cell technologies, such as nickel-metal hydride battery cells, sodium ion battery cells, lead-acid battery cells, and/or ultracapacitor cells. For example, in a nickel-metal hydride battery cell, the anode **208** may be formed from a hydrogen-absorbing alloy and the cathode **212** may be formed from a nickel oxide-hydroxide. In the example of a nickel-metal hydride battery cell, the electrolyte **210** may be formed from an aqueous potassium hydroxide in one or more examples.

[0067] The battery cell **120** may be implemented as a lithium sulfur battery cell in one or more other implementations. For example, in a lithium sulfur battery cell, the anode **208** may be formed at least in part from lithium, the cathode **212** may be formed from at least in part form sulfur, and the electrolyte **210** may be formed from a cyclic ether, a short-chain ether, a glycol ether, an ionic liquid, a super-saturated salt-solvent mixture, a polymer-gelled organic media, a solid polymer, a solid inorganic glass, and/or other suitable electrolyte materials.

[0068] In various implementations, the anode **208**, the electrolyte **210**, and the cathode **212** of FIG. **2**C can be packaged into a battery cell housing having any of various shapes, and/or sizes, and/or formed from any of various suitable materials. For example, battery cells **120** can have a cylindrical, rectangular, square, cubic, flat, pouch, elongated, or prismatic outer shape. As depicted in FIG. **2**D, for example, a battery cell such as the battery cell **120** may be implemented as a cylindrical cell. In the example of FIG. **2**D, the battery cell **120** includes a cell housing **215** having a cylindrical outer shape. For example, the anode **208**, the electrolyte **210**, and the cathode **212** may be rolled into one or more substantially cylindrical windings **221**. As shown, one or more windings **221** of the anode **208**, the electrolyte **210**, and the cathode **212** (e.g., and/or one or more separator layers such as separator layer **220**) may be disposed within the cell housing **215**. For example, a separator layer may be disposed between adjacent ones of the windings **221**. However, the cylindrical cell implementation of FIG. **2**D is merely illustrative, and other implementations of the battery cells **120** are contemplated.

[0069] For example, FIG. **2**E illustrates an example in which the battery cell **120** is implemented as a prismatic cell. As shown in FIG. **2**E, the battery cell **120** may have a cell housing **215** having a right prismatic outer shape. As shown, one or more layers of the anode **208**, the cathode **212**, and the electrolyte **210** disposed therebetween may be disposed (e.g., with separator materials between the layers) within the cell housing **215** having the right prismatic shape. As examples, multiple layer of the anode **208**, electrolyte **210**, and cathode **212** can be stacked (e.g., with separator materials between each layer), or a single layer of the anode **208**, electrolyte **210**, and cathode **212** can be formed into a flattened spiral shape and provided in the cell housing **215** having the right prismatic shape. In the implementation of FIG. **2**E, the cell housing **215** has a relatively thick cross-sectional width **217** and is formed from a rigid material. For example, the cell housing **215** in the implementation of FIG. **2**E may be formed from a welded, stamped, deep drawn, and/or impact extruded metal sheet, such as a welded, stamped, deep drawn, and/or impact extruded aluminum sheet. For example, the cross-sectional width **217** of the cell housing **215** of FIG. **2**E may be as much as, or more than 1 millimeter (mm) to provide a rigid housing for the prismatic battery cell.

In one or more implementations, the first terminal **216** and the second terminal **218** in the prismatic cell implementation of FIG. **2**E may be formed from a feedthrough conductor that is insulated from the cell housing **215** (e.g., a glass to metal feedthrough) as the conductor passes through to cell housing **215** to expose the first terminal **216** and the second terminal **218** outside the cell housing **215** (e.g., for contact with an interconnect structure **200** of FIG. **2**B). However, this implementation of FIG. **2**E is also illustrative and yet other implementations of the battery cell **120** are contemplated.

[0070] For example, FIG. **2**F illustrates an example in which the battery cell **120** is implemented as a pouch cell. As shown in FIG. **2**F, one or more layers of the anode **208**, the cathode **212**, and the electrolyte **210** disposed therebetween may be disposed (e.g., with separator materials between the layers) within the cell housing **215** that forms a flexible or malleable pouch housing. In the implementation of FIG. **2**F, the cell housing **215** has a relatively thin cross-sectional width **219**. For example, the cell housing **215** in the implementation of FIG. **2**F may be formed from a flexible or malleable material (e.g., a foil, such as a metal foil, or film, such as an aluminum-coated plastic film). For example, the cross-sectional width **219** of the cell housing **215** of FIG. **2**F may be as low as, or less than 0.1 mm, 0.05 mm, 0.02 mm, or 0.01 mm to provide flexible or malleable housing for the pouch battery cell. In one or more implementations, the first terminal **216** and the second terminal **218** in the pouch cell implementation of FIG. **2**F may be formed from conductive tabs (e.g., foil tabs) that are coupled (e.g., welded) to the anode **208** and the cathode **212** respectively, and sealed to the pouch that forms the cell housing **215** in these implementations. In the examples of FIGS. **2**C, **2**E, and **2**F, the first terminal **216** and the second terminal **218** are formed on the same side (e.g., a top side) of the battery cell **120**. However, this is merely illustrative and, in other implementations, the first terminal **216** and the second terminal **218** may formed on two different sides (e.g., opposing sides, such as a top side and a bottom side) of the battery cell **120**. The first terminal **216** and the second terminal **218** may be formed on a same side or difference sides of the cylindrical cell of FIG. **2**D in various implementations.

[0071] In one or more implementations, a battery module **115**, a battery pack **110**, a battery unit, or any other battery may include some battery cells **120** that are implemented as solid-state battery cells and other battery cells **120** that are implemented with liquid electrolytes for lithium-ion or other battery cells having liquid electrolytes. One or more of the battery cells **120** may be included a battery module **115** or a battery pack **110**, such as to provide an electrical power supply for components of the vehicle **100**, the building **180**, or any other electrically powered component or device. The cell housing **215** of the battery cell **120** can be disposed in the battery module **115**, the battery pack **110**, or installed in any of the vehicle **100**, the building **180**, or any other electrically powered component or device.

[0072] FIG. **3** illustrates a cross-sectional side view of the battery pack **110**, the cross section taken along the line A-A of FIG. **2**A, in accordance with one or more implementations of the subject technology. As shown in FIG. **3**, the battery pack **110** may include an energy volume enclosure **205** having a lid **277**. As shown, a space **333** may be defined by the top **257** and the sidewalls **259** of the lid **277**, and one or more battery subassemblies, such as one or more battery modules **115**, may be disposed within the space **333**. In one or more implementations, the one or more battery subassemblies (e.g., battery modules **115**) and/or battery cells **120** may be attached to the lid **277** (e.g., to an interior surface of the top **257** of the lid **277**) and enclosed within the energy volume enclosure **205** (e.g., within the space **333**). For example, the energy volume enclosure **205** may also include a tray **302** attached to the lid **277**. For example, the tray **302** may form a bottom wall or bottom plate of the energy volume enclosure **205**, and may be configured to attach to the lid **277** to enclose the one or more battery subassemblies and/or battery cells **120** within the space **333** defined by the top **257** and the sidewalls **259** of the lid **277**.

[0073] As shown in FIG. **3**, the battery pack **110** may include a gap **308** (e.g., an air gap) between the tray **302** and the one or more battery subassemblies. In one or more implementations, the one or more battery subassemblies may each include one or more battery cells **120**, each of the one or more battery cells having a vent **310** disposed adjacent to and/or within the gap **308**. For example, one or more battery cells **120** in the one or more battery subassemblies may include a vent **310** on a bottom side (e.g., a side facing the tray **302**) of the battery cell. Providing the gap **308** between the tray **302** and the one or more battery subassemblies may help to ensure that the vents **310** of the battery cells **120** are free of blockages or obstruction (e.g., by potting material that may be present elsewhere in the battery pack **110**). In one or more implementations, the battery cells **120** may include one or more terminals (e.g., terminals **216** and **218**) at a top end of the cell (e.g., adjacent the lid **277**). The terminals of the battery cells may be electrically coupled (e.g., by one or more current collector assemblies (CCAs), and/or busbars) to one or more high voltage terminals (e.g., terminals **314**) at the top of the battery pack **110**.

[0074] As described in further detail hereinafter, the battery pack **110** may also include a potting material **309** in an interstitial space between two or more battery cells **120** of each of the one or more battery subassemblies, and in a space between the one or more battery subassemblies and the lid **277**. The gap **308** between the tray **302** and the one or more battery subassemblies may be free of the potting material **309**.

[0075] As shown in FIG. **3**, the lid **277** may include a flange **304** (e.g., along the bottom edge of a sidewall formed by the lid **277**). The tray **302** may include a flange **306** that is mounted to the flange **304**. As shown, the sidewall structures **271** may be located on or above the flange **304**. Terminals **314** (e.g., high voltage terminals) may be provided for connecting electrical circuitry in the modular electrical component assembly **290** to the voltage generated by the battery cells **120** in the energy volume enclosure **205** (e.g., by electrically coupling the modular electrical component assembly **290** to one or more battery subassemblies within the energy volume **207** via the lid **277**, when the modular electrical component assembly **290** is mechanically coupled to the lid **277**).

[0076] FIG. **4** illustrates the battery pack **110** at two stages of assembly of the battery pack **110**. In this example, battery subassemblies (e.g., battery modules **115** and/or other groups of battery cells **120**) may be lowered into the lid **277** (e.g., as indicated by arrows **400**) while the lid **277** is in an inverted (e.g., “holding water”) position. For example, the lid **277** may be flipped such that the top of the lid **277** is the lowermost part of the lid **277** and an bottom opening in the lid **277** faces upward to allow the battery subassemblies to be lowered into the lid. As illustrated by arrow **402**, once the battery subassemblies and/or the battery cells **120** have been attached to the lid **277** (e.g., using one or more fasteners, using an adhesive, and/or by providing the potting material **309** into the lid **277** while the lid is in the inverted position and the battery subassemblies are held in contact with the lid **277** by gravity, and allowing the potting material to cure), the lid **277** with the battery subassemblies attached thereto may be flipped so that the top of the lid is at the top, and the opening of the lid is at the bottom.

[0077] As illustrated by the rear view of the battery pack **110** shown in FIG. **5**, the tray **302** may then be attached (e.g., by one or more fasteners **501**, such as bolts) to the lid **277**. For example, multiple bolts around the periphery of the energy volume enclosure **205** may be provided for attaching the tray **302** to the lid **277**. In one or more implementations, a skid plate **500** may be attached to the battery pack **110** (e.g., to the tray **302**, such as by one or more fasteners **503**, such as bolts). FIG. **5** also shows how the modular electrical component assembly **290** may be mounted to the top of the lid **277** when the lid **277** is an un-inverted (e.g., upright) position. FIG. **6** illustrates a rear view of the battery pack **110** with the tray **302** and the skid plate **500** attached to the lid **277** (e.g., to the bottom of the lid) and the modular electrical component assembly **290** mounted to the top of the lid **277**. In one or more implementations, the tray **302** and/or the skid plate **500** may be fastened to the lid **277** while the lid is in the inverted position (e.g., prior to flipping the lid with the battery subassemblies attached thereto), to allow the fasteners **501** and/or **503** to be fastened while the fasteners are pointing down, to aid in manufacturing ease. In the final orientation shown in FIG. **5** (e.g., the orientation in which the battery pack **110** may be attached to a vehicle **100**), serviceability of the battery pack **110** may be improved by providing a bottom stack of parts (e.g., the tray **302** and/or the skid plate **500**) that can be removed and replaced without having to do work on high voltage components. In one or more implementations, the tray **302** and/or the skid plate **500** may be formed from materials having higher strength and/or quality than the material of the lid **277**, as the lid to which the battery subassemblies are attached may be isolated from exposure to road abuse.

[0078] FIG. **7** illustrates a top view of an example battery subassembly, implemented as a battery module **115** (e.g., battery module **115**B) including multiple battery cells **120**. In one or more implementations, the battery subassembly may include a carrier **700** in which the battery cells **120** can be placed prior to insertion of the battery subassembly into the lid **277**. In one or more implementations, the carrier **700** may be formed from an insulating material, such as plastic. As shown in FIG. **7**, the carrier **700** may be provided with one or more datuming features, such as a two-way datum **702** at one end of the carrier **700** and a four-way datum **704** at an opposing end of the carrier **700**. In one or more implementations, the two-way datum **702** and/or the four-way datum **704** may also, or alternatively, be provided in a current collector assembly (CCA) of the battery subassembly. In the example of FIG. **7**, the battery subassembly (e.g., battery module **115**) is shown in a top-down view in a module pick position (e.g., positioned for lifting and inserting into the inverted lid **277**), such as after the flipping the battery module **115** after insertion of the battery cells **120** in the carrier **700**. In one or more implementations, a module lift assist may pin to the two-way datum **702** and/or the four-way datum **704** (e.g., on the carrier **700** and/or the CCA).

[0079] FIG. **8**A illustrates a top view of the lid **277** in the inverted position, after one battery module **115** has been lowered into the lid (e.g., between two longitudinal members **312** that run from the front to the rear of the lid **277**). As shown, the lid **277** may include spaces **800** (e.g., module bays), within the space **333**, for additional battery modules **115** (e.g., two additional battery modules **115**) or other battery subassemblies. FIG. **8**B illustrates how a battery module **115** (or other battery subassembly or group of battery cells **120**) may be assembled in a first orientation, and flipped (e.g., into an inverted configuration for the battery module) for insertion (e.g., lowering) into a space **800** in the inverted lid **277** (e.g., prior to flipping of the lid with the installed battery module **115** after the battery module **115** has been secured to the lid).

[0080] FIG. **9** illustrates a cross-sectional view of an enlarged portion of the battery pack **110** of FIG. **8**A. As illustrated by FIG. **9**, each of one or more battery subassemblies (e.g., battery modules **115**) may include a current collector assembly **902**. For example, the current collector assembly **902** may include multiple tabs welded to multiple battery cells **120**, in order to collect the current from the battery cells **120** for output from the battery pack **110**. As shown in FIG. **9**, the current collector assembly (CCA) **902** may include a flange **906** and/or a standoff **908**. The standoff **908** may define a spacing between the one or more battery subassemblies and the lid **277**. In one or more other implementations, the standoff **908** may be formed separately from the CCA **902** (e.g., on the lid **277**, or as a separate component of the battery pack **110**). In one or more implementations, the standoff **908** and/or the flange **906** may provide datum features (e.g., a primary datum feature for setting the location(s) of the battery subassembly (ies) (e.g., battery modules **115**) within the lid **277**. As shown in FIG. **9**, a seal **904** may extend around the terminals **314**, and may be compressed (e.g., by the weight of the battery subassembly while the lid **277** is in the inverted position), against an internal surface of the lid **277**. In this way, the battery pack **110** can be assembled with the lid **277** in a “holding water” (inverted) orientation so that the top of the battery modules **115** touch down on the lid assembly (e.g., an assembly including the lid **277**, the longitudinal members **312**, the sidewall structures **271**, and/or other battery pack features or components).

[0081] As illustrated by FIG. **9**, once the battery cells **120** (e.g., of a battery subassembly, such as a battery module **115**) have been inserted into the inverted lid **277**, a potting material (e.g., potting material **309** of FIG. **3**) may be provided (e.g., poured, dispensed, or otherwise provided, as indicated by arrows **907**) into the interstitial spaces **900** between the battery cells **120** (e.g. between two or more battery cells of each of the one or more battery subassemblies), and into a space **901** (e.g., defined by the standoffs **908**) between the one or more battery subassemblies (e.g., the battery cells **120**) and the lid **277**. Because the potting material is provided into the lid **277** while the lid is in the inverted position of FIG. **9** (e.g., and the potting material is thus pulled downward toward the top of the lid **277** and the CCA **902**, and into the space **901**, by gravity), the gap **308** (see, e.g., FIG. **3**) between the tray **302** and the one or more battery subassemblies may remain free of the potting material (e.g., without providing any particular components to block the flow of the potting material into the gap **308**, since the potting material naturally flows downward away from the gap **308**). For example, this orientation may allow the potting material **309** to be dispensed into the pack without fouling the cell vents and without a component to protect vent channels from being filled with potting material.

[0082] FIG. **10** illustrates various aspects of module X/Y datuming of the battery subassembly to the enclosure formed by the lid **277**. For example, a module load lift assist may be designed with selective compliance and centering, such as to limit re-alignment loads passed through CCA datums. As shown in FIG. **10**, one or more datums may be provided for aligning the battery modules **115** for insertion into the lid **277**. As examples, one or more datums **1000** may be provided on one or more of the longitudinal members **312**, one or more datums **1002** may be provided on one or more of the outer longitudinal members **1001**, and/or one or more datums **1004** may be provided on the flange **304** of the lid **277**. In one or more implementations, datums may also, or alternatively, be provided on a palette fixture **1006** to which the lid **277** may be mounted during assembly of the battery pack **110**. For example, the palette fixture **1006** may include one or more two-way datums **1008** at a first end thereof, and one or more four-way datums **1010** at an opposing second end thereof. In one or more implementations, a module load lift assist that lifts the battery module **115** (e.g., and that is aligned to the battery module **115** using the datum **702** and the datum **704** of FIG. **7**) may be aligned and/or indexed to the datums **1000**, **1002**, **1004**, **1008**, and/or **1010**, to align the battery module **115** for insertion into the lid **277** that is in the inverted position.

[0083] FIG. **11** illustrates a top view of the tray **302** in accordance with one or more implementations. As shown in FIG. **11**, the tray **302** may include one or more ribs **1100** (e.g., to form stiff beam elements on the tray panel). The tray **302** may also include a feature **1102** (e.g., a discontinuity) between adjacent pairs of the ribs **1100**. FIG. **12** illustrates an enlarged cross-sectional view of a portion of the tray **302**, taken along the line B-B of FIG. **11**. As shown in FIG. **12**, the feature **1102** may be configured to interface with a longitudinal member **312** of the energy volume enclosure **205** (e.g., of the lid **277**) to fluidly isolate a first region **1201** (e.g., a space **800** in which a first battery module **115** or other group of battery cells **120** is disposed) within the energy volume enclosure **205** from a second region **1202** (e.g., another space **800** in which a second battery module **115** or other group of battery cells **120** is disposed) within the energy volume enclosure **205**. As shown, a bolt **1200** may be configured to secure a metal-to-metal contact between the feature **1102** and the bottom end of the longitudinal member **312**. Providing this isolation achieved by interrupting the tray panel ribs **1100** at the longitudinal beams such that there is a metal-to-metal interface, the battery pack **110** may be configured to block the flow of air and/or debris from one module bay to another, including in the event of a thermal runaway caused by a cell or module failure.

[0084] As described herein, in one or more implementations, the battery pack **110** may also include a skid plate **500** attached to an outer surface of the tray **302**. In one or more implementations, the skid plate **500** may include one or more features that interface with the ribs **1100** of the tray **302**. For example, as illustrated in FIG. **13** (e.g., a cross-sectional view with the cross section taken along a longitudinal member **312**), the tray **302** may be attached to the longitudinal member **312** of the energy volume enclosure **205** by one or more fasteners **1300**, and the skid plate **500** may include one or more ribs **1302** positioned within a space defined by one of the ribs **1100** of the tray

302. For example, the one or more fasteners **1300** may include one or more t-nuts and corresponding bolts, and/or other types of nuts, bolts, screws, welds or other fasteners. Use of t-nuts as the fasteners **1300** may provide additional load distribution and/or stability for the connection between the tray **302** and the longitudinal member **312**. For example, the skid plate **500** may include cross car ribs **1302** inside tray panel ribs **1100** (e.g., extending into a space or recess formed by an undulation in the tray **302**, the undulation corresponding to one of the ribs **1100**). In one or more implementations, the ribs **1100** of the tray **302** may alternatively be implemented as crowned ribs (e.g., that protrude into corresponding recesses in the longitudinal member) to increase stiffness of ribs **1100**. In one or more implementations, the battery pack **110** may include an additional layer **1304** (e.g., a layer of aluminum, carbon fiber, glass fiber, or other sheet of material) between the tray **302** and the skid plate **500**. In one or more implementations, attaching the battery subassemblies and/or cells to the lid **277** (e.g., rather than the tray **302**) may be allow the ribs **1100** extend further outward toward the edges of the tray **302** and toward the location of a vehicle rocker of a vehicle in which the battery pack **110** is installed.

[0085] FIG. **14** illustrates a side perspective view of a portion of the battery pack **110**, in which the sidewall structures **271** can be seen. As shown, the sidewall structures **271** may include various impact absorption features mounted along a sidewall **1407** (e.g., one of the sidewalls **259** of the lid **277**) of the energy volume enclosure **205**. As described herein (see, e.g., FIG. **2**A), the energy volume enclosure **205** for the battery pack **110** may include a front end **267** and a rear end **269**. As shown in FIG. **14**, a sidewall **1407** that extends along a direction between the front end **267** and the rear end **269** may include impact absorption features. As shown, the battery pack **110** may include a first set impact absorption features (e.g., sidewall structures **271**) on an external surface of the sidewall **1407**. As shown, the first set of impact absorption features may include at least one beam (e.g., beam **1400** or beam **1404**) attached to an outer surface of the lid **277** and extending in a direction that runs between the front end **267** and the rear end **269**. For example, the at least one beam may include: a first beam **1404** attached (e.g., welded) to the outer surface of the lid **277** at a top of the sidewall **1407**, and a second beam **1400** attached (e.g., welded) to a flange **304** at a bottom of the lid **277**. As shown, the first set of impact absorption features may also include one or more reinforcement members **1402** attached to, and spaced apart along, the second beam **1400**. The sidewall structures **271** of FIG. **14** may provide an outer longitudinal section of the battery pack **110** for beam stiffness in GB crush and energy absorption in side pole impact tests.

[0086] FIG. **15** illustrates a cross-sectional view of a portion of the battery pack **110**, in which additional details of the sidewall structures **271** can be seen. As shown, a space **1501** may be provided between the beam **1404** and the reinforcement members **1402** (e.g., for mounting of cables, such as electrical, hydraulic, and/or thermal cables, that run between the front and the rear of a vehicle). As shown, in addition to the first set of impact absorption features (e.g., the beam **1400**, the beam **1404**, and the reinforcement members **1402**) on the external surface of the sidewall **1407**, the battery pack **110** may also include a second sent of impact absorption features on, and/or interior to, an internal surface of the sidewall **1407**. For example, the second set of impact absorption features may include a set of bulkheads **1500** that are attached to an outer longitudinal member **1001** and positioned interior to the lid **277**. For example, FIG. **16** illustrates a perspective view of the outer longitudinal member **1001** with multiple bulkheads **1500** attached (e.g., welded) thereto. As shown, the bulkheads **1500** may be spaced apart along a length of the outer longitudinal member **1001**. As shown, an additional bulkhead **1600** may be attached (e.g., welded) to the outer longitudinal member **1001** at an end of the outer longitudinal member **1001**. FIG. **17** illustrates a perspective view of the beam **1400** with multiple reinforcement members **1402** attached (e.g., welded) thereto.

[0087] FIG. **18** illustrates additional details of the battery pack **110** of FIG. **15** implemented in a vehicle **100**. As shown in FIG. **18**, the battery pack **110**, including the bulkheads **1500** attached to the outer longitudinal member **1001** and interior to the lid **277**, the beams **1400** and **1404** welded to

the outer surface of the lid **277**, and the reinforcement members **1402** attached (e.g., welded) to the beam **1400**, may be positioned interior to one or more vehicle body structures, such as a body rocker assembly **1800** and/or a body structure **1802**, of a vehicle (e.g., vehicle **100**). In the event of an impact to the vehicle **100** (e.g., to side of the vehicle **100**), the body rocker assembly **1800** and/or the body structure **1802** may transfer some or all of a force of the impact to the beam **1404**, the beam **1400**, and/or the reinforcement members **1402**. The beam **1404**, the beam **1400**, and/or the reinforcement members **1402** may distribute some or all of the force along the beams **1400** and/or **1404**, and/or may deform to absorb some or all of the force. In one or more use cases, some of the force may be transferred to the lid **277**, the bulkheads **1500**, and/or the outer longitudinal member **1001**. The lid **277**, the bulkheads **1500**, and/or the outer longitudinal member **1001** may distribute some or all of the force along the beams outer longitudinal member **1001** and/or **1404**, and/or may deform to absorb some or all of the force. Although the features of FIGS. **14-18** a described in connection with one sidewall **259** of the lid **277**, it is appreciated that the sidewall structures of **271** and bulkheads **277** may be repeated on an opposing sidewall **259** of the lid **277**.
[0088] As illustrated by FIGS. **15-18**, the battery pack **110** may be provided with sidewall features, including the outer longitudinal member **1001** (e.g., attached to the battery modules **115**), the beam **1400**, the beam **1404**, the reinforcement members **1402**, and/or the bulkheads **1500**, that include stiff beam members that allow a crush of outer portions of the pack to happen without loading modules and/or cells. Using the side structures described herein, the battery pack **110** may act as a load path for side impacts to the vehicle, without transferring the impact force to the battery cells. The long side members disclosed herein (e.g., the outer longitudinal members **1001**, the beam(s) **1404**, and/or the beam(s) **1400**) may provide torsional and bending stiffness to the battery pack **1110** and improve modal performance of the battery pack **110** and/or the vehicle **100**. In various implementations, any commodity, such as stampings, plastic inserts, and/or or extrusions can be used inside of the battery pack **110** pack as bulkheads **1500** for energy absorption and/or stiffness. In various implementations, any materials, such as steel, aluminum, or reinforced plastic can be used with the same structure concept. The side structure design described herein can be scaled up or down based on an expected magnitude of a load and based on vehicle mass.
[0089] In one or more implementations, the battery pack **110** may form a structural component of the vehicle **100**. For example, the vehicle **100** may be provided without a structural frame that runs from the front to the rear of the vehicle. For example, the vehicle body and front and/or rear subframes may be mounted to the battery pack **110**. Providing the sidewall structures **271** as described herein may allow the battery pack **110** to perform side-impact protection functions previously performed by a vehicle frame, while reducing the weight and materials of the vehicle (e.g., by removing the frame), thereby meeting vehicle safety standards while increasing the range of the vehicle.
[0090] FIG. **19** illustrates a flow diagram of an example process **1900** that may be performed for assembling a battery pack, in accordance with implementations of the subject technology. For explanatory purposes, the process **1900** is primarily described herein with reference to the battery pack **110** of FIGS. **1**A-**18**. However, the process **1900** is not limited to the battery pack **110**, and one or more blocks (or operations) of the process **1900** may be performed by or with one or more other structural components of other devices, systems, or battery assemblies or subassemblies. Further for explanatory purposes, some of the blocks of the process **1900** are described herein as occurring in serial, or linearly. However, multiple blocks of the process **1900** may occur in parallel. In addition, the blocks of the process **1900** need not be performed in the order shown and/or one or more blocks of the process **1900** need not be performed and/or can be replaced by other operations.
[0091] As illustrated in FIG. **19**, at block **1902**, a battery subassembly (e.g., a battery module **115**, such as a battery module **115**B, **115**D, or **115**F) may be lowered (e.g., as indicated by arrows **400** of FIG. **4**) into a lid (e.g., lid **277**) of an enclosure (e.g., energy volume enclosure **205**) of a battery pack (e.g., battery pack **110**) while the lid is in an inverted position (e.g., as illustrated in FIGS. **4**

and **8**B). In the inverted position of the lid, an opening **431** spanning a bottom of the lid may face upward to allow objects to be lowered into the lid.

[0092] At block **1904**, the battery subassembly may be attached to the lid. Attaching the battery subassembly to the lid may include attaching a carrier for one or more battery cells (e.g., battery cells **120**) to the lid and/or attaching the battery cells **120** to the lid.

[0093] At block **1906**, a potting material may be provided into an interstitial space (e.g., interstitial space **900**) between two or more battery cells (e.g., battery cells **120**) in the battery subassembly. In one or more implementations, attaching the battery subassembly to the lid at block **1904** may include attaching the battery subassembly to the lid using the potting material (e.g., by allowing the potting material to flow into and cure in the space(s) between the battery subassembly and the lid, and thereby attach the battery subassembly to the lid). In one or more other implementations, attaching the battery subassembly to the lid may include attaching the battery subassembly to the lid using one or more fasteners (e.g., bolts, screws, clips, etc.), using one or more welds, using an adhesive, or any other suitable attachment mechanism. For example, in one or more implementations, attaching the battery subassembly to the lid may include providing a potting material into the lid containing the battery subassembly, and allowing the potting material to cure in one or more interstitial spaces between battery cells of the battery subassembly, and between the lid and a carrier (e.g., carrier **700**) and/or a CCA (e.g., CCA **902**). Attaching the battery subassembly to the lid may also include allowing the potting material to flow downward, due to gravity, toward the lid **277** and into the interstitial spaces between battery cells of the battery subassembly, and between the lid and the carrier (e.g., carrier **700**) and/or the CCA (e.g., CCA **902**).

[0094] At block **1908**, the lid having the battery subassembly attached thereto may be flipped (e.g., as illustrated in FIG. **4**). Flipping the lid may be performed before or after a tray (e.g., tray **302**) and/or one or more other layers (e.g., a skid plate **500** and/or a layer **1304**) have been attached to the lid, to cover the opening in the bottom of the lid that is upward facing when the lid is in the inverted position.

[0095] For example, at block **1910**, the battery subassembly may be enclosed within the energy volume enclosure (e.g., within the space **333**) by attaching a tray (e.g., tray **302**) of the energy volume enclosure to the lid of the energy volume enclosure. For example, the tray may be attached to the lid of the energy volume enclosure using one or more bolts or other fasteners. In one or more implementations, the tray **302** and/or a skid plate **500** may be attached to the lid to close the energy volume enclosure prior to flipping the lid having the battery subassembly attached thereto.

[0096] In one or more implementations, the process **1900** may also include mechanically coupling a modular electrical component assembly (e.g., modular electrical component assembly **290**) to the lid, and electrically coupling the battery subassembly to the modular electrical component assembly via the lid (e.g., using one or more connectors that pass through one or more respective openings in the top of the lid). Mechanically and electrically coupling the modular electrical component assembly to the energy volume enclosure may include bolting an enclosure of the modular electrical component assembly to the lid of the energy volume enclosure, and electrically coupling one or more high voltage terminals (e.g., terminals **314**) on the energy volume enclosure to electrical circuitry within the modular electrical component assembly. In one or more implementations, the process **1900** may also include (e.g., at block **1912**) providing the battery pack in a vehicle (e.g., vehicle **100**).

[0097] In one or more implementations, lowering the battery subassembly into the lid may include lowering the battery subassembly into the lid may include while the battery subassembly is in an inverted configuration for the battery subassembly. In one or more implementations, lowering the battery subassembly into the lid may include loading the battery subassembly into the lid dry (e.g., with no potting in the battery subassembly). The potting can then be provided, at block **1906**, at a pack level, and to fill in the inter-cell spaces and the connection to the lid **277** at the same time. For example, providing the potting material at block **1906** may include dispensing a potting material that cures as an adhesive through the cells **120**. As the potting material flows in, it may fill into the bottom of the inverted lid **277** and cure to hold the battery subassembly in place. In this way, the potting material can form an adhesive the bears loads including for shock and vibe crash durability, or the like. By providing a robust attachment between the battery cells **120** and the lid **277** of the battery pack **110** in this way, the mechanical performance of the cells may contribute to the stiffness of the battery pack **110** and/or a vehicle **100** in which the battery pack is installed. For example, the moment of inertia of the entire top of the energy volume enclosure **205** can be increased by attaching the battery cells **120** and/or other battery subassembly structures at the top of the battery pack **110** (e.g., attached to the lid **277** as described herein).

[0098] Attaching the battery subassemblies to the lid **277** as described herein may help to reduce the tolerances and/or the number of materials in the Z-stack of the battery pack **110**. For example, the location of the terminals **314** and/or associated busbars extending through the lid **277** from the battery subassemblies, relative to the mounting location of interfacing terminals of the modular electrical component assembly **290** on the top of the lid **277**, may be controlled more precisely by lowering the battery modules **115** into the inverted lid **277** as shown and described herein (e.g., in contrast with mounting the battery subassemblies to a bottom of the energy volume enclosure and then installing the lid above the battery subassemblies). Attaching the battery subassemblies to the lid **277** as described herein may also help to optimize the amount of potting material **309** (e.g., a foaming or other curing potting material) used in a battery pack **110**, such as by optimizing the thickness of the spaces to be filled with that potting material and using gravity to move the potting material to, rather than away from, desired potting locations at the tops of the cells and/or the CCA. Thus, attaching the battery subassemblies to the lid **277** as described herein may reduce the number of components for sealing and limiting flow of potting material into places (e.g., gap **308**) where the presence of potting material is undesirable. Accordingly, attaching the battery subassemblies to the lid **277** as described herein may provide additional benefits in terms of reduced cost, weight, and/or manufacturing complexity, while providing improvements in structural performance.

[0099] Aspects of the subject technology can help improve the reliability and/or range of electric vehicles. This can help facilitate the functioning of and/or proliferation of electric vehicles, which can positively impact the climate by reducing greenhouse gas emissions.

[0100] A reference to an element in the singular is not intended to mean one and only one unless specifically so stated, but rather one or more. For example, “a” module may refer to one or more modules. An element proceeded by “a,” “an,” “the,” or “said” does not, without further constraints, preclude the existence of additional same elements.

[0101] Headings and subheadings, if any, are used for convenience only and do not limit the invention. The word exemplary is used to mean serving as an example or illustration. To the extent that the term include, have, or the like is used, such term is intended to be inclusive in a manner similar to the term comprise as comprise is interpreted when employed as a transitional word in a claim. Relational terms such as first and second and the like may be used to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions.

[0102] Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0103] A phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list. The phrase “at least one of” does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, each of the phrases “at least one of A, B, and C” or “at least one of A, B, or C” refers to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

[0104] It is understood that the specific order or hierarchy of steps, operations, or processes disclosed is an illustration of exemplary approaches. Unless explicitly stated otherwise, it is understood that the specific order or hierarchy of steps, operations, or processes may be performed in different order. Some of the steps, operations, or processes may be performed simultaneously. The accompanying method claims, if any, present elements of the various steps, operations or processes in a sample order, and are not meant to be limited to the specific order or hierarchy presented. These may be performed in serial, linearly, in parallel or in different order. It should be understood that the described instructions, operations, and systems can generally be integrated together in a single software/hardware product or packaged into multiple software/hardware products.

[0105] In one aspect, a term coupled or the like may refer to being directly coupled. In another aspect, a term coupled or the like may refer to being indirectly coupled.

[0106] Terms such as top, bottom, front, rear, side, horizontal, vertical, and the like refer to an arbitrary frame of reference, rather than to the ordinary gravitational frame of reference. Thus, such a term may extend upwardly, downwardly, diagonally, or horizontally in a gravitational frame of reference.

[0107] The disclosure is provided to enable any person skilled in the art to practice the various aspects described herein. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology. The disclosure provides various examples of the subject technology, and the subject technology is not limited to these examples. Various modifications to these aspects will be readily apparent to those skilled in the art, and the principles described herein may be applied to other aspects.

[0108] All structural and functional equivalents to the elements of the various aspects described throughout the disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f), unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for”.

[0109] Those of skill in the art would appreciate that the various illustrative blocks, modules, elements, components, methods, and algorithms described herein may be implemented as hardware, electronic hardware, computer software, or combinations thereof. To illustrate this interchangeability of hardware and software, various illustrative blocks, modules, elements, components, methods, and algorithms have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application. Various components and blocks may be arranged differently (e.g., arranged in a different order, or partitioned in a different way) all without departing from the scope of the subject technology.

[0110] The title, background, brief description of the drawings, abstract, and drawings are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the

scope or meaning of the claims. In addition, in the detailed description, it can be seen that the description provides illustrative examples and the various features are grouped together in various implementations for the purpose of streamlining the disclosure. The method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The claims are hereby incorporated into the detailed description, with each claim standing on its own as a separately claimed subject matter.
[0111] The claims are not intended to be limited to the aspects described herein, but are to be accorded the full scope consistent with the language of the claims and to encompass all legal equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirements of the applicable patent law, nor should they be interpreted in such a way.

Claims

1. An apparatus, comprising: a battery pack, comprising: a lid that forms a top and a plurality of sidewalls of an energy volume enclosure of the battery pack; one or more battery subassemblies enclosed within a space defined by the top and the plurality of sidewalls of the lid; and a tray attached to the lid to enclose the one or more battery subassemblies within the space defined by the top and the plurality of sidewalls of the lid.
2. The apparatus of claim 1, wherein the tray forms a bottom wall of the energy volume enclosure and is configured to attach to the lid using a plurality of fasteners disposed around a periphery of the energy volume enclosure to enclose the one or more battery subassemblies within the space defined by the top and the plurality of sidewalls of the lid, and wherein the battery pack further comprises an air gap between the tray and the one or more battery subassemblies.
3. The apparatus of claim 2, wherein the one or more battery subassemblies each comprise one or more battery cells, and wherein each of the one or more battery cells have a vent disposed adjacent the air gap.
4. The apparatus of claim 2, wherein each of the one or more battery subassemblies is attached to the lid, and wherein each of the one or more battery subassemblies comprises a current collector assembly having a standoff that defines a spacing between the one or more battery subassemblies and the lid.
5. The apparatus of claim 4, wherein the battery pack further comprises a potting material in an interstitial space between two or more battery cells of each of the one or more battery subassemblies, and in a space between the one or more battery subassemblies and the lid.
6. The apparatus of claim 5, wherein the air gap between the tray and the one or more battery subassemblies is free of the potting material, and wherein the potting material attaches the one or more battery subassemblies to the lid.
7. The apparatus of claim 2, wherein the tray comprises: a plurality of ribs; and a feature between adjacent pairs of the plurality of ribs, the feature configured to interface with a longitudinal member of the energy volume enclosure to fluidly isolate a first region within the energy volume enclosure from a second region within the energy volume enclosure.
8. The apparatus of claim 7, wherein the battery pack further comprises a skid plate attached to an outer surface of the tray.
9. The apparatus of claim 8, wherein the tray is attached to the longitudinal member of the energy volume enclosure by one or more fasteners, and wherein the skid plate comprises one or more ribs positioned within a space defined by one of the ribs of the tray.
10. The apparatus of claim 1, further comprising a plurality of impact absorption features mounted along one of the plurality of sidewalls of the lid.

11. The apparatus of claim 1, wherein the apparatus comprises a vehicle, and wherein the battery pack is implemented in the vehicle.

12. An enclosure for a battery pack, the enclosure comprising: a front end; a rear end; a sidewall extending between the front end and the rear end; a first plurality of impact absorption features on an external surface of the sidewall; and a second plurality of impact absorption features on an internal surface of the sidewall.

13. The enclosure of claim 12, further comprising: a lid; and an outer longitudinal member extending from the front end to the rear end, wherein the second plurality of impact absorption features comprise a plurality of bulkheads that are attached to the outer longitudinal member and positioned interior to the lid.

14. The enclosure of claim 13, wherein the first plurality of impact absorption features include at least one beam attached to an outer surface of the lid, the at least one beam extending in a direction that runs between the front end and the rear end.

15. The enclosure of claim 14, wherein the at least one beam comprises: a first beam attached to the outer surface of the lid at a top of the sidewall; and a second beam attached to a flange at a bottom of the lid.

16. The enclosure of claim 15, wherein the first plurality of impact absorption features comprise a plurality of reinforcement members attached to, and spaced apart along, the second beam.

17. A method, comprising: lowering a battery subassembly into a lid of an energy volume enclosure of a battery pack while the lid is in an inverted position; attaching the battery subassembly to the lid; providing a potting material into an interstitial space between two or more battery cells in the battery subassembly; flipping the lid having the battery subassembly attached thereto; and enclosing the battery subassembly within the energy volume enclosure by attaching a tray of the energy volume enclosure to the lid of the energy volume enclosure.

18. The method of claim 17, wherein lowering the battery subassembly into the lid comprises lowering the battery subassembly into the lid while the battery subassembly is in an inverted configuration of the battery subassembly, the method further comprising providing the battery pack in a vehicle.

19. The method of claim 17, wherein attaching the battery subassembly to the lid comprises attaching the battery subassembly to the lid with the potting material.

20. The method of claim 19, further comprising mechanically coupling a modular electrical component assembly to the lid and electrically coupling the modular electrical component assembly to the battery subassembly via the lid.
